A Graph-Based Hyperspectral Change Detection Framework Using Difference Augmentation and Progressive Reconstruction With Limited Labels

Bin Yang, *Member, IEEE*, Xinwei Cheng, Wei Chen, and Xin Ye

***Abstract*— Identifying land cover changes based on hyperspectral images (HSIs) has been a research hotspot in the field of remote sensing (RS). In recent years, deep-learning (DL)-based change detection (CD) methods have advanced the development of this subject due to their powerful feature representation capabilities. However, it is difficult for these methods to mine changed information between bi-temporal HSIs with limited labels. To overcome this limitation, we propose a graph-based hyperspectral CD framework using difference augmentation and progressive reconstruction (ARCD), which enhances the recognition ability of changes in HSIs with limited labels. This framework consists of three components: 1) a dual-branch multiscale dynamic graph convolutional network (DMGCN) subnetwork, which is developed to emphasize the changed information and learn global features of HSIs at various scales; 2) a difference augmentation feature fusion (DAFF) module, which is designed to fuse spectral–spatial augmentation information and the difference information to accurately capture discriminative features for the changes between bi-temporal HSIs; and 3) a progressive contextual information attention reconstruction (PCAR) module, which is proposed to focus on key information in the context, and progressively reconstruct multilevel features to reduce semantic gaps between different scale features. ARCD not only enhances the representation ability of changed features but also alleviates the demand for HSI labels. We test the performance of ARCD on four hyperspectral datasets. Quantitative and qualitative results reveal that it outperforms some state-of-the-art (SOTA) methods with limited labels.**

***Index Terms*— Change detection (CD), difference augmentation, graph convolutional network (GCN), hyperspectral images (HSIs), limited labels, progressive reconstruction.**

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I. Introduction

CHANGE detection (CD) aims to identify changes in the land surface by comparing remote sensing (RS) images captured at different times of the same region [1]. This technique has been widely applied in various fields, including disaster assessment [2], natural resource surveys [3], and meteorological monitoring [4]. The primary data for CD are multispectral images, high-resolution images, and synthetic aperture radar images [5]. Compared to these data, hyperspectral images (HSIs), which contain hundreds of contiguous spectral bands [6], [7], [8], can provide richer spectral information of ground objects and thus give a significant advantage in CD task [9].

HSI CD has long been a focus in the field of RS, and numerous methods have been proposed. These methods can be categorized into three types: algebra-based methods, transformation-based methods, and classification-based methods [10], [11]. The algebra-based methods rely on mathematical modeling of the algebraic properties of changed and unchanged objects. Widely used mathematical techniques include regression, difference, ratio, and change vector analysis (CVA) [12]. CVA is one of the most representative algorithms that quantifies the extent of change by analyzing the Euclidean distance between bi-temporal images and dividing the changed and unchanged regions with a threshold value. On the basis of CVA, some improved methods have been proposed, including robust CVA (RCVA) [13] and compressed CVA (C^2VA) [14]. The algebra-based methods are relatively easy to use, but they are sensitive to illumination changes and noise [15]. The transformation-based methods focus on transforming the feature space of HSIs, and subsequently, comparing the transformed features from different phases to identify changed regions [16]. One of the most widely used methods is principal component analysis (PCA) [17], which extracts significant features from high-dimensional data and clusters them using the K-means method [18]. In addition, many methods are also based on transformation, including the slow feature analysis (SFA) [19], the deep belief network [20], and the iteratively reweighted multivariate alteration detection (IR-MAD) [21]. The transformation-based methods can suppress some of the noise, but they are unable to make full use of the spectral information in HSIs [22]. The classification-based methods, such as the support vector machine (SVM)

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[23], and the maximum likelihood classifier (MLD) [24], utilize classifiers to categorize pixels or regions within images to detect changes [25]. However, the performance of these methods is highly dependent on the choice of classifiers and may lead to cumulative errors [26]. The aforementioned methods often rely on manually designed features, which have limited ability to represent complex land cover changes in hyperspectral data.

In recent years, deep-learning (DL) technology has achieved great success in the field of CD [27], [28]. DL-based CD methods that can automatically learn feature representations and process large hyperspectral data of complex scenes [29], [30]. Convolutional neural networks (CNNs), as a representative DL framework, have produced numerous outstanding contributions [31]. For example, Wang et al. [32] proposed an end-to-end 2-D CNN framework, in which a mixed-affinity matrix was introduced, and then CNNs were employed to identify changed regions. Considering temporal dependence within bi-temporal HSIs, Qu et al. [33] presented a multilevel encoder–decoder attention network (ML_EDAN), in which the long short-term memory (LSTM) subnetwork analyzed the temporal information from the multilevel spatial–spectral features extracted by CNNs. Song et al. [34] proposed a novel network, named CSANet, in which cross-temporal interaction symmetric attention integrated difference features from different time phases. To capture subtle changes in HSIs, Luo et al. [35] proposed a multiscale diff-changed feature fusion network (MSDFFN), which learned refined changed features across different scales. However, the above CNN-based methods are limited by their receptive field, which makes it difficult to model long-term information [36]. The transformer-based methods excel in capturing long-term dependencies [37], [38], and they have also shown remarkable effectiveness in HSI CD. Wang et al. [39] proposed a triplet transformer framework, which extracted HSI features using a "branch-backbone-branch" structure to improve the CD performance. To make full use of the features of HSIs, Wang et al. [40] presented a novel transformer method, in which the spectral–spatial transformer encoded spectral–spatial sequences and the temporal transformer captured temporal correlation. Zhang et al. [41] introduced a novel cascade transformer that effectively extracts and integrates spatial–spectral–temporal features through its temporal–spatial transformer and spectral-awareness transformer. Despite that both the CNN-based and the transformer-based methods have achieved satisfactory performance, they demand a large number of labels, otherwise, there is a risk of overfitting in the network.

Notably, acquiring a significant number of labels for HSIs is a time-consuming and laborious task [42], [43]. To overcome the problem of an insufficient number of manual labels, there have been efforts to explore methods that can operate with limited labels, for example, Cai et al. [44] proposed a novel two-step learning method, which can accurately classify images using a small number of training labels. Sun et al. [45] proposed a large kernel spectral–spatial attention network, which achieves excellent performance with only about 0.1% training labels. In addition, some unsupervised methods have been proposed. For example, Shi et al. [46] addressed this problem by generating pseudo-labels. They presented a collaborative analysis framework, in which a self-perception network and a cross-task knowledge transfer mechanism were proposed to enable collaborative analysis across multiple HSI datasets. In contrast to [46], there are some unsupervised methods that do not require the generation of pseudo-labels. For example, the method proposed by Lei et al. [47] trained the network with the idea of adversarial learning for unsupervised CD. Yang et al. [48] proposed an untrained framework using randomly initialized models (RICD), which extracted discriminative features and highlighted changed pixels. The performance of these methods is often limited by factors, including pseudo-labels or training strategies. These limitations drove us to explore new frameworks for HSI CD.

Recently, the graph convolutional network (GCN) has attracted increasing attention for its ability to train networks in a semi-supervised manner and achieve excellent performance even with limited labels [49], [50], [51]. In the field of HSI CD, Qu et al. [52] were the first to apply GCN to the research, and they presented a dual-branch GCN framework, in which a difference amplification module was designed to highlight changed pixels. Wang et al. [53] applied the graph attention network (GAT) to the HSI CD task. They proposed a network that simultaneously extracted pixel-level features using CNNs and superpixel-level features using GAT, and made full use of the strengths of both feature extraction methods to complement each other. Lin and Lin [54] combined DL and convex optimization to implement HSI CD. A Q-quadratic feature extractor was designed to optimize preliminary CD results obtained using a simple GAT-based network. Nevertheless, these methods do not pay attention to the difference in information between bi-temporal images during the process of graph convolution. Additionally, they only consider the single-scale feature of HSIs when constructing graph nodes, lacking the ability to utilize contextual spatial–spectral information in HSIs.

Based on the above analysis, due to the semi-supervised training characteristics of GCNs, we have identified significant potential in fully extracting and integrating changed features of different scales between bi-temporal HSIs using GCN with limited labels. In this article, we propose a graph-based hyperspectral CD framework using difference aenugmtation and progressive reconstruction (ARCD) for HSI CD. First, a dual-branch multiscale dynamic GCN (DMGCN) subnetwork is designed, which focuses on the changed information and dynamically extracts global features at different scales from HSIs. Then, a difference augmentation feature fusion (DAFF) module is designed to accurately capture discriminative features for the changes between bi-temporal HSIs. Finally, a progressive contextual information attention reconstruction (PCAR) module is proposed to fully utilize the rich multiscale information and contextual information in HSIs and progressively achieve the feature reconstruction. Due to the combination of these modules, ARCD can achieve powerful feature extraction capabilities even with limited labels. The main contributions are summarized as follows.

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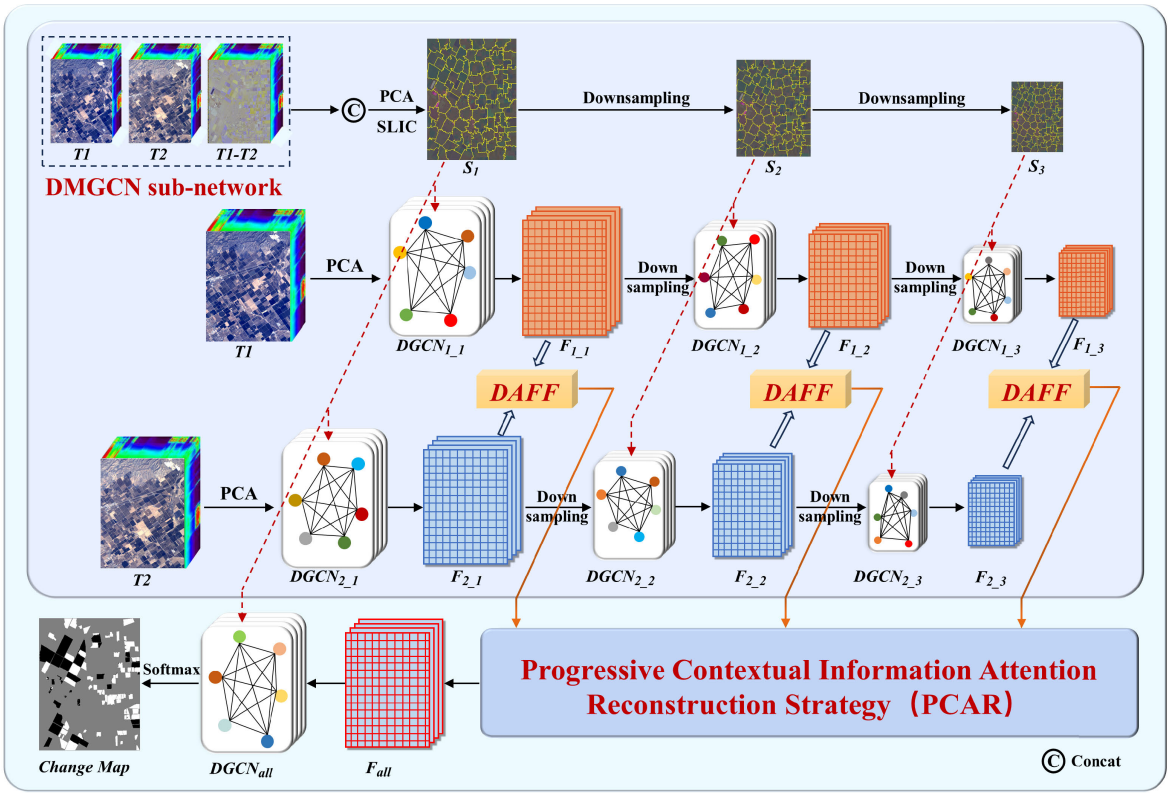


Fig. 1. Overall network framework of the proposed ARCD.

1) A novel graph-based hyperspectral CD framework, called ARCD, is proposed, which dynamically learns discriminative features of HSIs to enhance the recognition of changed surfaces even with limited labels.
2) A DAFF module is designed, which fuses spectral–spatial augmentation information and difference information to extract discriminative textural features in bi-temporal HSIs.
3) A PCAR module is proposed, which captures key information in the context, and achieves multilevel difference feature reconstruction in a progressive manner to reduce semantic gaps between different scale features.

The remainder of this article is divided into three sections. Section II describes the details of the proposed ARCD. Section III presents the experimental results and analysis. Finally, Section IV provides a conclusion for this article.

II. Proposed Method

In this section, the framework of ARCD is presented in Fig. 1. It consists of the DMGCN subnetwork, DAFF module, and PCAR module. First, the original HSIs are sent to the DMGCN subnetwork to dynamically extract global features at different scales from HSIs. Then, the DAFF module combines spectral–spatial augmentation information with difference information to highlight the changed features. Finally, the PCAR module progressively reconstructs multilevel features to fully utilize the rich contextual information in HSIs. In the following, we will provide a detailed description for each module of ARCD.

A. Dual-Branch Multiscale DMGCN Subnetwork

1) Framework of DMGCN: HSIs contain rich spatial–spectral information that is challenging to be adequately captured by a regular single-scale GCN. To tackle this concern, a DMGCN subnetwork, including multiscale graph node generation and DMGCN, is proposed, which focuses on the changed information and dynamically captures global information at different scales from HSIs. It is illustrated in Fig. 1.

The DMGCN subnetwork extracts features by aggregating graph nodes. Due to the abundance of pixels in HSIs, many graph-based methods treat superpixels as graph nodes [53], which significantly reduces the number of nodes. Moreover, knowing the label of a pixel within the superpixel determines the label of the corresponding node. However, most methods do not take into account the unique characteristics of the CD task, and they do not focus on the changed regions between bi-temporal HSIs during the process of graph convolution. Therefore, we propose a new multiscale graph node generation method, which utilizes multiscale segmentation masks to guide the generation of graph nodes.

In the DMGCN subnetwork, the bi-temporal HSIs are donated as $T1, T2 \in \mathbb{R}^{H\times W\times C}$, and their difference image is donated as $T3 \in \mathbb{R}^{H\times W\times C}$, where H and W donate the height and width of HSIs, and C donates the number of spectra. First, to pay attention to the changed information between bi-temporal images, we concatenate $T1$, $T2$, and $T3$ along the channel dimension. Then, the concatenated image is reduced to three dimensions using PCA [17]. Finally, it is segmented

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using simple linear iterative clustering (SLIC) to generate mask S_1 based on the spatial similarity among pixels [55]. The entire process is defined as

$$S_1 = f_{\text{slic}}(\text{PCA}(\text{Concat}[T1; T2; T3])) \tag{1}$$

where Concat[;] is concatenation operation along the channel dimension, PCA$(\cdot)$ represents PCA operation, and f_{slic} donates SLIC operation.

To obtain multiscale masks, we perform a downsampling operation on the obtained mask. Notably, in the process of graph construction and feature updating, although the size of the mask is different, the number of regions divided is always unchanged. In other words, the number of graph nodes remains consistent across the different scales of the GCN. The generation process of multiscale masks is described as

$$S_i = \text{Downsampling}(S_{i-1}). \tag{2}$$

Based on the generated multiscale masks, the overall process of the DMGCN subnetwork is as follows. First, to reduce the amount of computation, PCA operation is employed to retain 99.9% of the original information in the input HSIs. Second, the first-scale $\text{DGCN}_{i_1}(i = 1, 2)$ (DGCN is described in the next section) are constructed using the mask S_1 obtained by the SLIC algorithm and the pixel-level features obtained by PCA operation. Then, the updated pixel-level features $F_{i_1}(i = 1, 2)$ are obtained by aggregating the features of graph nodes. Next, downsampling operation is applied to both S_1 and $F_{i_1}(i = 1, 2)$ for generating the feature map and mask S_2 of the same size. Subsequently, the second-scale $\text{DGCN}_{i_2}(i = 1, 2)$ is reconstructed to obtain the updated feature $F_{i_2}(i = 1, 2)$. Similarly, we can obtain the output features $F_{i_3}(i = 1, 2)$ for the third-scale $\text{DGCN}_{i_3}(i = 1, 2)$. By dynamically capturing multiscale features of HSIs, the network enhances its capabilities for perceiving and representing hyperspectral data, thereby gaining a more comprehensive understanding of HSIs.

2) Dynamic GCN: DMGCN has demonstrated excellent performance in feature extraction of hyperspectral data [52], [53]. During the model learning process, it dynamically updates the adjacency matrix based on the features of the data. In other words, the relationships between various graph nodes are redefined with each layer of the GCN. The process of DGCN is as follows.

First, based on S_1, the input pixel-level features $P = \{p_i\}_{i=1}^{n}$ are divided into N superpixel regions $S = \{s_j\}_{j=1}^{N}$, where p_i denotes the ith pixel in HSIs, $n = H \times W$ is the number of all pixels, $s_j = \{p_k^j\}_{k=1}^{Y_j}$ denotes the jth superpixel region, p_k^j donates the kth pixel in s_j, and Y_j denotes the number of pixels contained in the superpixel region s_j. Furthermore, the average value of pixels within each superpixel region is calculated as a feature value $V = \{V_j\}_{j=1}^{N}$, which is then used to construct the nodes of DMGCN. In other words, the obtained N graph nodes are formulated as

$$V_j = \frac{1}{Y_j} \sum_{k=1}^{Y_j} p_k^j \tag{3}$$

where V_j donates the value of the jth graph node.

Then, a learnable similarity relationship between graph nodes is established by calculating Mahalanobis distance during the network training process. It is defined as

$$\mathcal{D}(V_a, V_b) = \sqrt{(V_a - V_b)^{\top} M (V_a - V_b)} \tag{4}$$

where $\mathcal{D}(\cdot)$ donates the similarity relationship between graph nodes, V_a and V_b donate any two graph nodes, and $M = W_d W_d^T$ donates symmetric positive semidefinite matrix, where W_d is a learnable matrix. The global adjacency matrix $A \in \mathbb{R}^{N \times N}$ is constructed as

$$A_{ab} = e^{-\gamma(\mathcal{D}(V_a, V_b))^2} \tag{5}$$

where γ is a hyperparameter.

Finally, the pixel feature updating strategy is employed using adjacency matrices, and then graph nodes are updated based on pixel features, as described in [56]. Taking the jth graph node as an example, it is calculated as

$$Z^{\text{out}} = \theta\left(\left(A_{jj} \times Z^{\text{in}} + \sum_{i=1}^{n} A_{ij} \times V_i\right) \cdot W\right), \quad i \neq j \tag{6}$$

where Z^{in} and Z^{out} represent the input and output pixel-level features in the jth superpixel, respectively. W donates a trainable weight, and $\theta(\cdot)$ is the activation function.

B. DAFF Module

In the field of CD, a widely used approach for obtaining difference features involves directly subtracting the bi-temporal features. However, direct subtraction may result in the loss of important information [57], which can significantly undermine the performance of CD. To address this issue and obtain more accurate difference features with limited labels, we propose a DAFF module, which aims to effectively fuse spectral–spatial augmentation information and difference information for improving the localization of the detailed and textural features in HSIs. The details of DAFF are shown in Fig. 2.

First, the concatenated features of size $2C \times H \times W$ and the difference features of size $C \times H \times W$ are obtained based on the bi-temporal features, which are defined as

$$F_{\text{cat}} = \text{Concat}[F_{T1}; F_{T2}] \tag{7}$$

$$F_{\text{diff}} = |F_{T1} - F_{T2}| \tag{8}$$

where $F_{T1} \in \mathbb{R}^{C \times H \times W}$ and $F_{T2} \in \mathbb{R}^{C \times H \times W}$ donate the output features at times $T1$ and $T2$, respectively. $|\cdot|$ denotes the absolute value operation.

Then, to obtain refined changed features, spectral augmentation, and spatial augmentation operations are applied to the concatenated feature F_{cat} in two branches, resulting in spectral augmentation weights of size $C \times 1 \times 1$ and spatial augmentation weights of size $1 \times H \times W$. Next, the spectral augmentation weights and spatial augmentation weights obtained are multiplied element-wise with the difference features F_{diff}, respectively. This process highlights the changed

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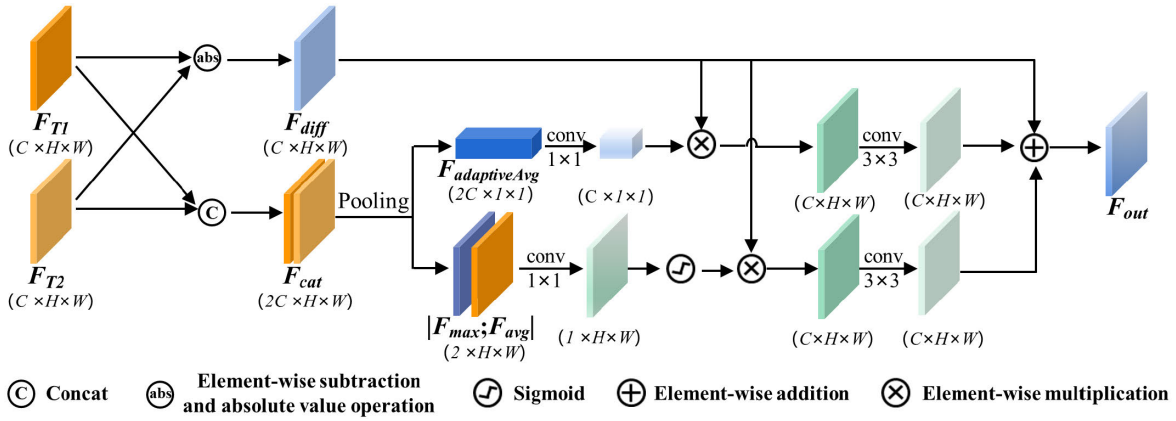


Fig. 2. DAFF module.

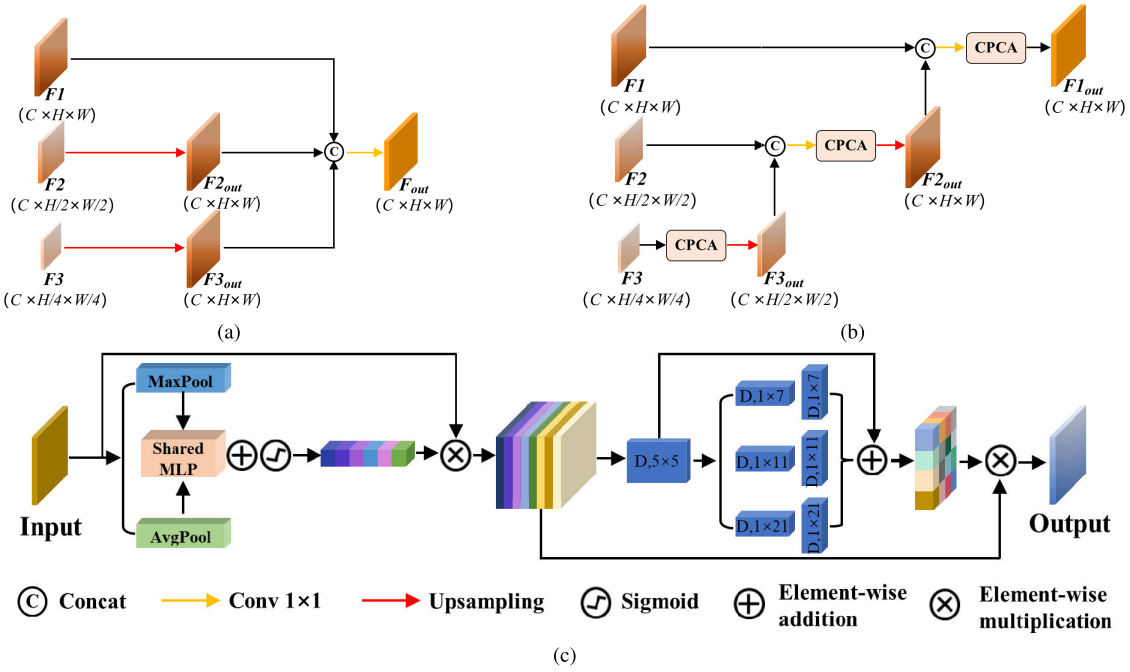


Fig. 3. Comparison of different methods for contextual information reconstruction. (a) DIR. (b) PCAR. (c) CPCA mechanism.

information and suppresses the unchanged information at the spectral and spatial levels. The process is described as

$$F_{\text{spectral}} = f_{\text{conv}}^{3\times3}\left(f_{\text{conv}}^{1\times1}(\text{AdaAvgpool}(F_{\text{cat}})) \cdot F_{\text{diff}}\right) \tag{9}$$

$$F_{\text{spatial}} = f_{\text{conv}}^{3\times3}\left(\sigma\left(f_{\text{conv}}^{1\times1}\left(\text{Concat}\left[\text{Maxpool}(F_{\text{cat}}); \text{Avgpool}(F_{\text{cat}})\right]\right)\right) \cdot F_{\text{diff}}\right) \tag{10}$$

where F_{spectral} and F_{spatial} denote spectral and spatial augmentation results, respectively. AdaAvgpool$(\cdot)$ denotes the adaptive average pooling operation, and MaxPool$(\cdot)$ and AvgPool$(\cdot)$ represent max-pooling and average-pooling operations, respectively. $f_{\text{conv}}^{1\times1}$ is a convolutional layer with a filter size of 1×1, $f_{\text{conv}}^{3\times3}$ is a convolutional layer with a filter size of 3×3, and $\sigma(\cdot)$ donates sigmoid function.

Finally, the features of two branches are fused with the difference features to attain the final output result F_{out}. Guided by the spectral–spatial augmentation information, the final difference features can capture more prominent changed information, which can be represented as

$$F_{\text{out}} = F_{\text{spectral}} + F_{\text{spatial}} + F_{\text{diff}}. \tag{11}$$

C. PCAR Module

The information embedded in features at different levels is distinct [58]. Low-level features contain more detailed information, but their semanticity is poor, while high-level scale features are rich in semantic information but have lower spatial resolution. How to combine features of different levels for efficient reconstruction is crucial for the HSI CD task. A widely used method is direct information reconstruction (DIR), as shown in Fig. 3(a). In DIR, this method resizes the features at multiple scales in a specific manner (e.g., upsampling) to align with the spatial size of the final feature of the network. Subsequently, the concatenation operation is applied to obtain the ultimately reconstructed features. However, due

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to significant differences in detailed and semantic information among different hierarchical features, DIR fails to fully utilize feature information across adjacent levels.

To enhance the representation capability of multilevel difference features generated by the DAFF module to alleviate label requirements, we propose a PCAR module, which is shown in Fig. 3(b). Fig. 3(c) illustrates the channel prior convolutional attention (CPCA) mechanism [59], which dynamically allocates attention weights across both channel and spatial dimensions. As a result, the PCAR module effectively captures abundant attentional information within the context through the utilization of the CPCA mechanism and then progressively reconstructs multilevel features to reduce the semantic gap among features of different scales.

The working process of the PCAR module is as follows. First, the third-level features $F3$ perform the CPCA mechanism and upsampling operation to obtain features of the same size as the upper-level features. Next, the concatenation operation is applied to the obtained features and the upper-level features at the channel level. Then, a 1×1 convolution is utilized to reduce the channel number from $2C$ to C. Finally, the features are progressively reconstructed in the same manner as the upper-level feature until the final result of feature reconstruction is obtained. The whole feature reconstruction process is expressed as

$$\begin{cases} F3_{\text{out}} = \text{Up}(\text{CPCA}(F3)) \\ F2_{\text{out}} = \text{Up}\big(\text{CPCA}\big(f_{\text{conv}}^{1\times1}(\text{Concat}[F2;\, F3_{\text{out}}])\big)\big) \\ F1_{\text{out}} = \text{Up}\big(\text{CPCA}\big(f_{\text{conv}}^{1\times1}(\text{Concat}[F1;\, F2_{\text{out}}])\big)\big) \end{cases} \tag{12}$$

where $F1 \in \mathbb{R}^{C\times H\times W}$, $F2 \in \mathbb{R}^{C\times(H/2)\times(W/2)}$, $F3 \in \mathbb{R}^{C\times(H/4)\times(W/4)}$ donate three input features, $F1_{\text{out}} \in \mathbb{R}^{C\times H\times W}$, $F2_{\text{out}} \in \mathbb{R}^{C\times H\times W}$, $F3_{\text{out}} \in \mathbb{R}^{C\times(H/2)\times(W/2)}$ represent three reconstruction results, and CPCA$(\cdot)$ donates CPCA mechanism.

D. Loss Function

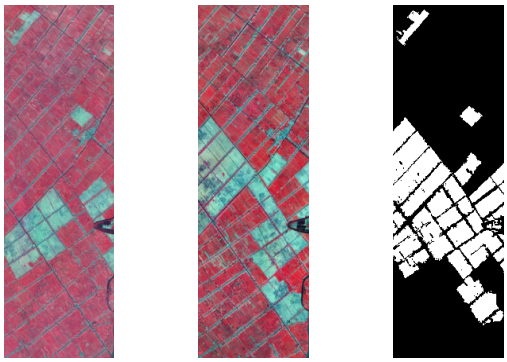
The choice of loss function is crucial for efficiently optimizing the proposed network. HSI CD is approached as a binary classification task, where the pixel values in the ground-truth (GT) are classified as either changed or unchanged. Therefore, we follow a standard cross-entropy loss function to train the proposed ARCD, as did in [52]

$$\mathcal{L}_{\text{ce}} = -\frac{1}{n}\sum_{i=1}^{n}\Big(X_i \log \widehat{X}_i + (1 - X_i)\log\Big(1 - \widehat{X}_i\Big)\Big) \tag{13}$$

where n is the number of labeled pixels, X and $\widehat{X}$ are the GT value and predicted value, respectively. $\mathcal{L}_{\text{ce}}$ represents the difference between X and $\widehat{X}$. In general, the smaller the value of $\mathcal{L}_{\text{ce}}$, the closer the change map is to GT.

III. Experimental Results and Analysis

In this section, comprehensive experiments are designed to demonstrate the effectiveness of the ARCD method. First, four HSI datasets and five evaluation criteria used in the experiments are introduced. Second, eight comparative methods are described, and a detailed explanation of the experimental setup is provided. The superiority of the ARCD method is confirmed through rigorous validation, which is supported by qualitative and quantitative experimental results. Then, ablation studies are performed to evaluate the effectiveness of DAFF and PCAR. Finally, followed by four discussions, we examine the impact of experimental parameter settings on the network.



(a) (b) (c)

Fig. 4. Pseudo-color images and GT of Farmland dataset. (a) HSI captured on May 3, 2006. (b) HSI captured on April 23, 2007. (c) GT (white: changed, black: unchanged).

A. Hyperspectral Datasets and Evaluation Criteria

Four hyperspectral datasets are used for comparative experiments, each dataset comprises two HSIs and a GT map. Before experiments, the acquired HSIs underwent rigorous preprocessing steps, including atmospheric and geometric correction. The first two datasets, "Farmland" [32] and "Hermiston" [61], were collected by the Earth Observing-1 (EO-1) Hyperion sensor [62], while the other two datasets, "Santa Barbara" [63] and "Bay Area" [63], were collected by the Airborne Visible InfraRed Imaging Spectrometer (AVIRIS) sensor.

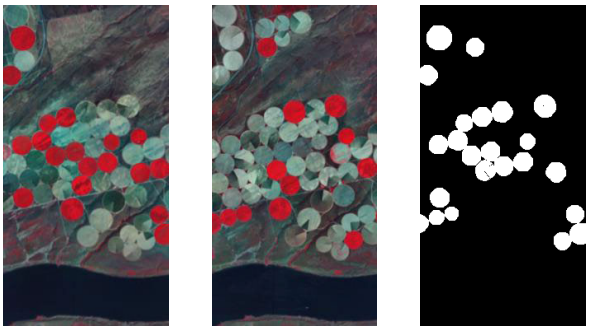
1) EO-1 Hyperion Datasets: EO-1 Hyperion sensor has a wavelength coverage of 0.4–2.5 μm. It acquires datasets with a spectral resolution of 10 nm and a spatial resolution of 30 m, containing 242 bands.

The Farmland dataset captures changes in the farmland near Yancheng, Jiangsu Province, China. The pseudo-color images are shown in Fig. 4(a) and (b). This dataset was acquired on May 3, 2006, and April 23, 2007. Each image has a spatial size of 450×140 pixels and contains 155 spectral bands. Fig. 4(c) shows the GT map of the Farmland dataset, which contains 44 723 changed pixels and 18 277 unchanged pixels.

The Hermiston dataset captures agricultural changes in the Hermiston, Oregon, USA. The pseudo-color images are shown in Fig. 5(a) and (b). This dataset was acquired on May 1, 2004, and May 8, 2007. Each image has a spatial size of 390×200 pixels and contains 198 spectral bands. Fig. 5(c) shows the GT map of the Hermiston dataset, which contains 9986 changed pixels and 68 014 unchanged pixels.

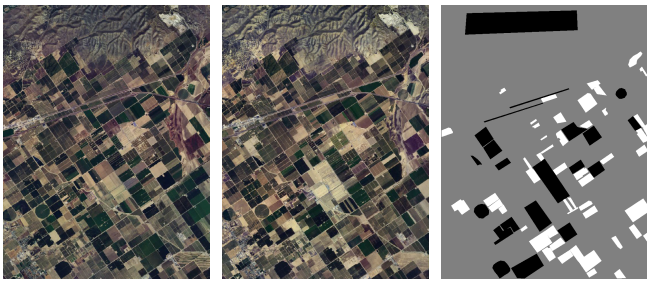
2) AVIRIS Datasets: AVIRIS sensor was developed by NASA's Jet Propulsion Laboratory and covers a wavelength range of 0.4–2.5 μm. It acquires datasets with a spectral resolution of 10 nm and a spatial resolution of 20 m, containing 224 bands.

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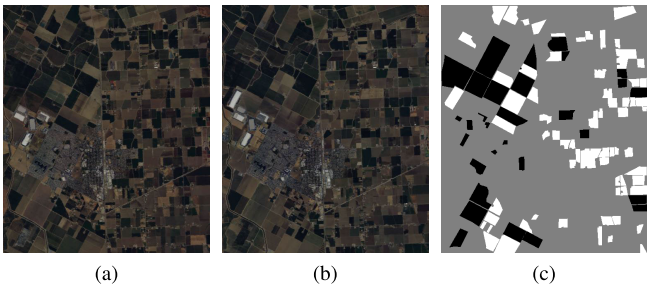
(a) (b) (c)

Fig. 5. Pseudo-color images and GT of Hermiston dataset. (a) HSI captured on May 1, 2004. (b) HSI captured on May 8, 2007. (c) GT (white: changed, black: unchanged).



(a) (b) (c)

Fig. 6. Pseudo-color images and GT of Santa Barbara dataset. (a) HSI captured in 2013. (b) HSI captured in 2014. (c) GT (white: changed, black: unchanged, gray: unmarked).



(a) (b) (c)

Fig. 7. Pseudo-color images and GT of Bay Area dataset. (a) HSI captured in 2013. (b) HSI captured in 2015. (c) GT (white: changed, black: unchanged, gray: unmarked).

The Santa Barbara dataset was acquired in 2013 and 2014 over the Santa Barbara region of California, USA. The pseudo-color images are shown in Fig. 6(a) and (b). Each image has a spatial size of 984 × 740 pixels and contains 224 spectral bands. Fig. 6(c) is the GT map of the Santa Barbara dataset, which contains 52 134 changed pixels, 80 418 unchanged pixels, and the remaining pixels are unmarked.

The Bay Area dataset was acquired in 2013 and 2015 over the Patterson City region of California, USA. The pseudo-color images are shown in Fig. 7(a) and (b). Each image has a spatial size of 600 × 500 pixels and contains 224 spectral bands. Fig. 7(c) is the GT map of the Bay Area dataset, which contains 39 270 changed pixels, 34 211 unchanged pixels, and the remaining pixels are unmarked.

To quantitatively assess the performance of the proposed ARCD, this article employs five commonly used evaluation criteria: overall accuracy (OA), kappa coefficient (Kappa) [64], Precision, Recall, and F1-score. OA represents the proportion of correctly predicted pixels to the total number of labeled pixels. Kappa measures the consistency between the change map and the GT map. Precision represents the proportion of correctly predicted changed pixels among the pixels predicted as changed. Recall represents the correctly predicted proportion of pixels that have actually changed. F1-score represents the harmonic average of Recall and Precision. The calculation of evaluation criteria is detailed as follows:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{14}$$

$$\text{Kappa} = \frac{\text{OA} - \text{PRE}}{1 - \text{PRE}} \tag{15}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{16}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{18}$$

$$\text{PRE} = \frac{(\text{TP} + \text{FN})(\text{TP} + \text{FP}) + (\text{TN} + \text{FN})(\text{TN} + \text{FP})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2} \tag{19}$$

where PRE is accidental consistency error, TP represents the number of pixels that are identified as changed in both GT map and the predicted results, TN donates the number of pixels that are both identified as unchanged, FP is the number of pixels that are labeled as unchanged in GT map but predicted as changed, and FN is the number of pixels that are labeled as changed in GT map but predicted as unchanged.

B. Compared Methods and Experimental Setup

In this article, eight state-of-the-art (SOTA) CD methods are selected to validate the advancement of ARCD method, including three unsupervised algorithms, namely CVA [12], SFA [19], and RICD [48], as well as five semi-supervised algorithms, namely GCN [49], ML-EDAN [33], CSANet [34], MDGCN [51], and MSDFFN [35].

1) *CVA:* CVA is a widely used CD method that quantifies the extent of change by utilizing the magnitude of vector differences between bi-temporal images, and changed and unchanged regions are determined through thresholding [12].
2) *SFA:* SFA is a method based on spatial transformation. It can capture changed information in time series data by learning a set of slowly changed features, thereby achieving CD in images [19].
3) *RICD:* RICD is a randomly initialized method based on spatial-channel enhancement. The difference between changed pixels and unchanged pixels is amplified by feature selection and spatial-channel enhancement strategies [48].
4) *GCN:* GCN is the first application of graph convolution in the field of DL, which updates node information by aggregating features from neighboring nodes for node

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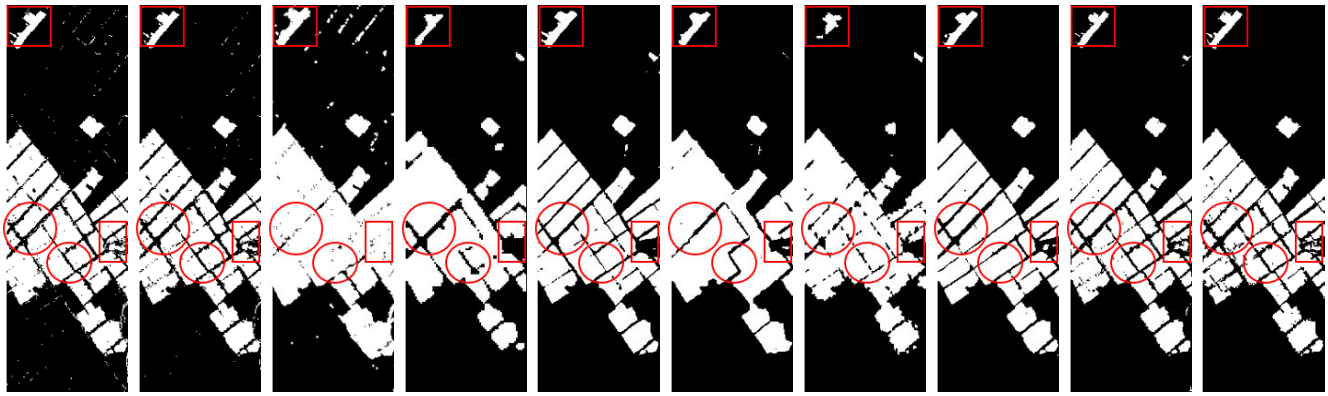


Fig. 8. Qualitative results of each CD algorithm on the Farmland dataset. (a) CVA. (b) SFA. (c) RICD. (d) GCN. (e) ML_EDAN. (f) CSANet. (g) MDGCN. (h) MSDFFN. (i) ARCD. (j) GT map.

classification. In our experiment, this network is applied to the HSI CD task [49].

5) *ML-EDAN:* ML-EDAN is an HSI CD method based on multilevel CNN and LSTM networks. The LSTM subnetwork analyzes and integrates the temporal information from multilevel features extracted by CNNs [33].
6) *CSANet:* CSANet is a cross-temporal feature interaction method that improves the recognition of changed regions by extracting spatial–spectral–temporal features from HSIs [34].
7) *MDGCN:* MDGCN is proposed based on DMGCN, which aggregates features from multiple neighborhood scales, thoroughly exploring the spatial–spectral correlations among multiscale features [51].
8) *MSDFFN:* MSDFFN is an HSI CD method based on multiscale difference feature fusion, which focuses on subtle changed features within each scale of bi-temporal features in HSIs to enhance the network's discriminative capability for fine features [35].

Due to the challenge of obtaining labels for HSI CD, the model's performance at a low sampling rate is of great importance. In our experiments, we randomly select 0.5% of both changed and unchanged pixels as training samples, while the remaining data is used for validation. The parameters of the comparative algorithms are set as specified in the original paper. The ARCD method is implemented using the PyTorch framework on an NVIDIA GTX 3090Ti GPU [65]. During the training process, we employ the Adam optimizer [66] to optimize the network. Considering that the EO-1 sensor datasets show concentrated regions of change, while the AVIRIS sensor datasets are more fragmented, we choose different experimental parameters. For EO-1 Hyperion datasets, we set the superpixel segmentation to 20 and the total number of training epochs to 200. The learning rate is set to 5e-5 and decayed by a factor of 0.6 every 150 epochs. For AVIRIS datasets, we set the superpixel segmentation to 150 and the total number of training epochs to 800. The learning rate is set to 5e-4 and decayed by a factor of 0.6 every 150 epochs.

C. Experimental Results

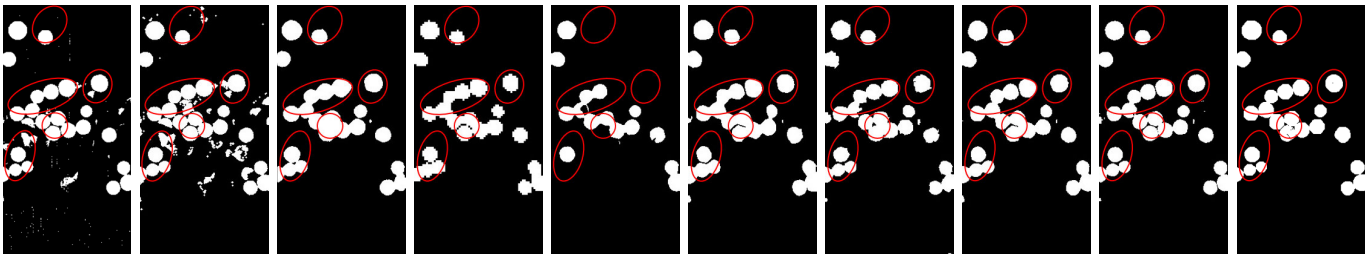
In this section, we analyze the quantitative and qualitative experimental results of each algorithm on four datasets separately. In quantitative analysis, the best results are highlighted in bold font, while the second-best results are underlined.

1) Farmland Dataset: The qualitative results of each algorithm on the Farmland dataset are shown in Fig. 8. While the unsupervised algorithms show the ability to detect the majority of changed regions, there are still limitations in accurately identifying certain detailed information. For instance, some unchanged pixels in the upper right and lower left corners of the change map are erroneously classified as changed. Among the semi-supervised algorithms, both the GCN and CSANet algorithms are capable of detecting the majority of changed regions. However, the obtained detection results exhibit a certain level of coarseness, and the edges of the changed regions are inflated. In contrast, the change maps generated by the ML_EDAN, MDGCN, and MSDFFN algorithms demonstrate a significant improvement in visual quality. This suggests that multiscale networks are more effective in capturing the spatial–spectral features in HSIs. Our proposed ARCD achieves the best visual results on the Farmland dataset, and the change map is the closest to the GT map.

The quantitative results of each algorithm on the Farmland dataset are presented in Table I. Except for the GCN algorithm, the semi-supervised algorithms achieve similar values across various evaluation metrics, indicating that they have comparable detection performance. In contrast, the two unsupervised algorithms, CVA and SFA, exhibit higher OA and Kappa values compared to the semi-supervised algorithms. Notably, our proposed ARCD achieves the maximum values in OA, kappa, Precision, and F1-score metrics, which are 97.13%, 93.05%, 94.81%, and 95.07%, respectively.

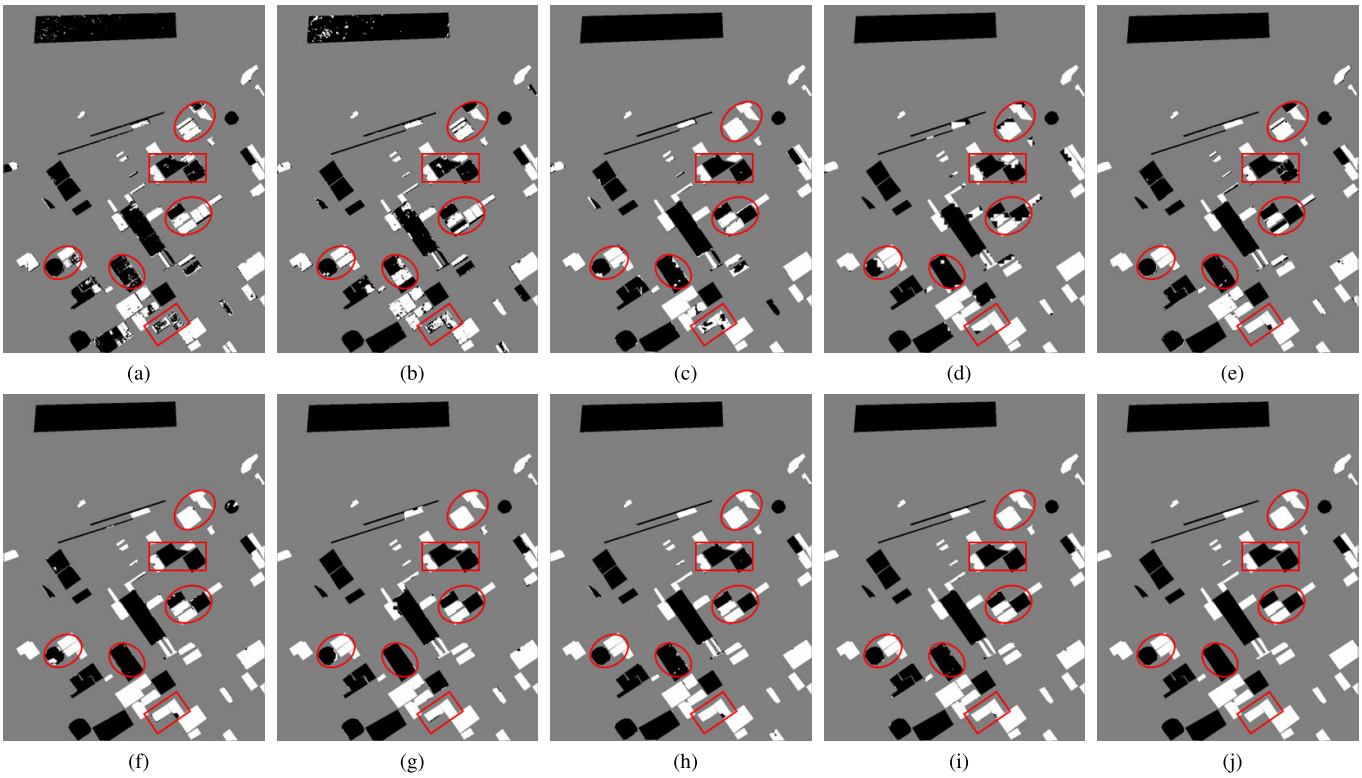
2) Hermiston Dataset: The qualitative results of each algorithm on the Hermiston dataset are shown in Fig. 9. Among the unsupervised algorithms, both the CVA and SFA algorithms exhibit some degree of false detection. In contrast, the visual performance of the RICD algorithm is improved. This suggests that the feature selection and spatial-channel enhancement strategies designed in the RICD algorithm effectively enhance the model's ability to capture changed features. Among the semi-supervised algorithms, the ML_EDAN algorithm exhibits a degree of missed detection, which could

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 9. Qualitative results of each CD algorithm on the Hermiston dataset. (a) CVA. (b) SFA. (c) RICD. (d) GCN. (e) ML_EDAN. (f) CSANet. (g) MDGCN. (h) MSDFFN. (i) ARCD. (j) GT map.



(a) (b) (c) (d) (e)

(f) (g) (h) (i) (j)

Fig. 10. Qualitative results of each CD algorithm on the Santa Barbara dataset. (a) CVA. (b) SFA. (c) RICD. (d) GCN. (e) ML_EDAN. (f) CSANet. (g) MDGCN. (h) MSDFFN. (i) ARCD. (j) GT map.

be attributed to the limited labels. The remaining comparative algorithms can learn the changed features across most regions, but they struggle to accurately identify the detailed edges of circular changed regions. On the whole, the visual results of our proposed ARCD are closely aligned with the GT map.

The quantitative results of each algorithm on the Hermiston dataset are presented in Table I. The Precision values for the CVA and SFA algorithms are 85.23% and 73.19%, respectively, indicating the presence of some falsely detected areas. The ML_EDAN algorithm achieves the Recall value of 63.40%, which indicates that many changed regions are undetected. The OA and kappa values achieved by the remaining comparative algorithms are similar to each other and consistent with the qualitative results. Our proposed ARCD achieves maximum values in the assessments of three comprehensive metrics: OA, Kappa, and F1-score, reaching 98.74%, 94.33%, and 95.05%, respectively.

3) Santa Barbara Dataset: The qualitative results of each algorithm on the Santa Barbara dataset are shown in Fig. 10. Among the unsupervised algorithms, the CVA and SFA algorithms generate noise in their results, while the RICD algorithm exhibits some degree of false detection. Among the semi-supervised algorithms, the change maps obtained generally outperform those of the unsupervised algorithms. However, the GCN and ML_EDAN algorithms misclassify many unchanged pixels within the circled regions. The CSANet, MDGCN, and MSDFFN algorithms all achieve relatively favorable detection results, but there are still some noisy points within the circled regions. It is evident that the visual map of ARCD is superior to the aforementioned algorithms.

The quantitative results of each algorithm on the Santa Barbara dataset are presented in Table I. The CVA and SFA algorithms exhibit poor detection performance with relatively low values across multiple metrics. In contrast, the RICD, GCN, and ML_EDAN algorithms achieve all metric values exceeding 90%, while the CSANet, MDGCN, and MSDFFN algorithms attain all metric values surpassing 96%, showcasing

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TABLE I
QUANTITATIVE CD RESULTS ON FOUR HYPERSPECTRAL DATASETS (%)

Dataset	Index	Unsupervised methods			Semi-supervised methods					
		CVA	SFA	RICD	GCN	ML_EDAN	CSANet	MDGCN	MSDFFN	ARCD
Farmland	OA	96.11	96.79	89.58	93.69	95.81	93.95	93.92	95.78	**97.13**
	Kappa	90.70	92.24	76.96	84.94	89.98	85.71	85.49	89.52	**93.05**
	Precision	91.21	93.98	74.11	86.87	90.72	86.16	87.90	90.09	**94.81**
	Recall	95.84	95.04	**98.55**	92.17	95.02	94.29	91.76	95.35	95.34
	F1-score	93.47	94.50	84.60	89.44	92.96	90.04	89.79	92.65	**95.07**
Hermiston	OA	97.08	94.81	98.18	97.98	94.99	98.04	98.37	98.38	**98.74**
	Kappa	87.42	79.25	92.15	90.96	73.77	91.57	92.78	92.91	**94.33**
	Precision	85.23	73.19	89.47	91.76	**96.22**	88.41	92.77	91.33	95.19
	Recall	93.34	93.83	97.28	92.49	63.40	**97.42**	94.68	96.52	94.92
	F1-score	89.10	82.23	93.20	92.13	76.43	92.70	93.71	93.85	**95.05**
Santa Barbara	OA	84.05	87.28	95.56	95.26	96.09	98.18	98.13	98.92	**99.11**
	Kappa	64.56	72.99	90.63	90.06	91.71	96.18	96.07	97.74	**98.14**
	Precision	94.39	86.50	96.27	94.29	97.89	97.59	98.63	98.73	**98.89**
	Recall	82.41	80.17	92.30	93.63	92.02	97.77	96.59	98.53	**98.86**
	F1-score	88.00	83.21	94.24	93.96	94.87	97.68	97.60	98.63	**98.87**
Bay Area	OA	86.89	77.34	93.38	94.74	95.15	96.78	97.83	97.27	**98.40**
	Kappa	73.90	55.07	86.76	89.39	90.27	93.56	95.66	94.52	**96.79**
	Precision	93.18	85.14	96.90	93.36	96.06	98.22	98.17	98.78	**99.02**
	Recall	81.43	69.78	90.50	97.05	94.82	95.72	97.76	96.07	**97.97**
	F1-score	86.91	76.70	93.59	95.17	95.43	96.96	97.97	97.41	**98.49**

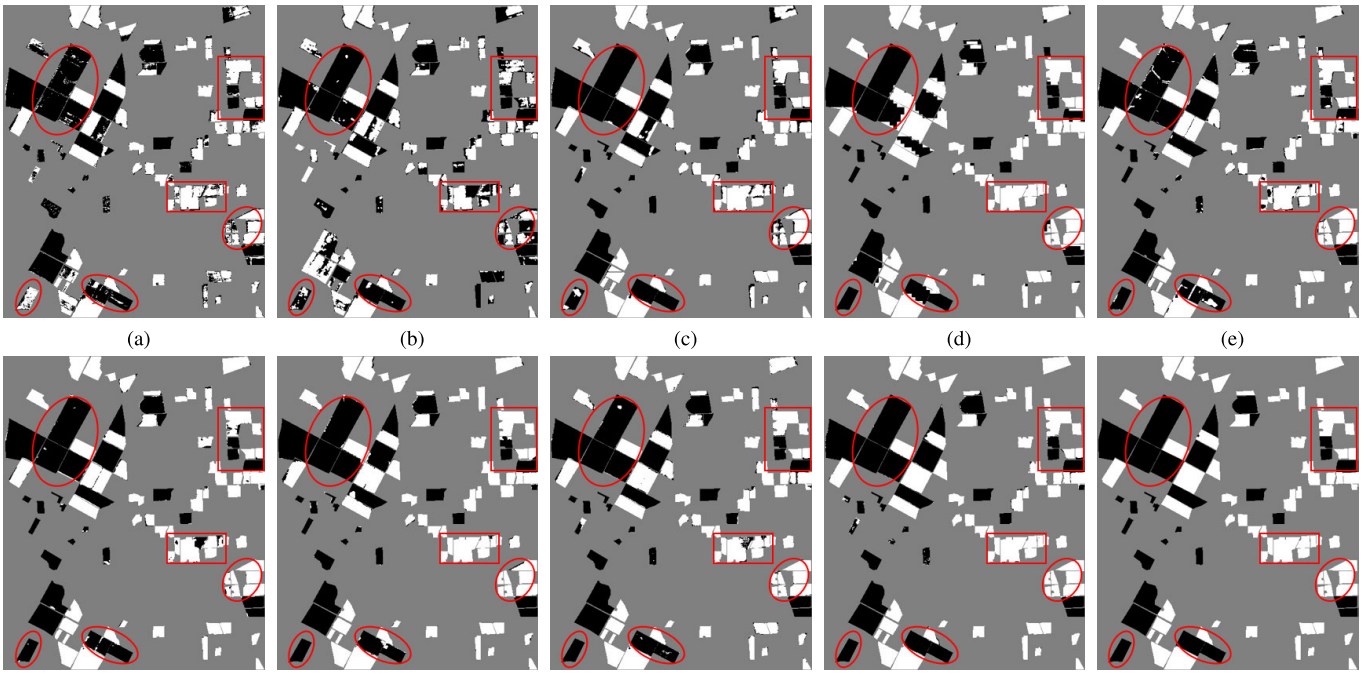


Fig. 11. Qualitative results of each CD algorithm on the Bay Area dataset. (a) CVA. (b) SFA. (c) RICD. (d) GCN. (e) ML_EDAN. (f) CSANet. (g) MDGCN. (h) MSDFFN. (i) ARCD. (j) GT map.

superior detection effectiveness consistent with qualitative results. Notably, our proposed ARCD achieves maximum values in all metrics, demonstrating its robust performance even under conditions of limited labels.

4) Bay Area Dataset: The qualitative results of each algorithm on the Bay Area dataset are shown in Fig. 11. Among the unsupervised algorithms, the CVA and SFA algorithms produce numerous noisy points, while the RICD algorithm identifies certain changed pixels as unchanged. This is attributed to the intricate distribution of changed materials on the surface, which poses a challenge to accurate identification by unsupervised algorithms. Among the semi-supervised algorithms, the change maps obtained by the GCN, ML_EDAN, and CSANet algorithms contain some pixels of false detection, particularly within the circled regions. Although the MDGCN and MSDFFN algorithms are effective in detecting most of the changed areas, crucial details within the circled regions are partially missing. It is obvious that our proposed ARCD produces visual results that are most similar to the GT map.

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TABLE II
RESULTS OF ABLATION EXPERIMENTS ON THE FARMLAND AND SANTA BARBARA DATASETS (%)

Dataset	Model	(1)	(2)	(3)	(4)
	DAFF	×	√	×	√
	DIR	√	√	×	×
	PCAR	×	×	√	√
Farmland	OA	95.67	96.21	96.75	**97.13**
	Kappa	89.34	90.28	92.12	**93.05**
	Precision	93.36	93.65	94.18	**94.81**
	Recall	90.16	92.71	94.65	**95.34**
	F1-score	91.73	93.18	94.42	**95.07**
Santa Barbara	OA	97.31	97.84	98.84	**99.11**
	Kappa	94.36	95.49	97.56	**98.14**
	Precision	96.66	96.97	98.72	**98.89**
	Recall	96.49	97.57	98.32	**98.86**
	F1-score	96.58	97.27	98.52	**98.87**

TABLE III
RESULTS OF DIFFERENT SEGMENTATION SCALES ON THE FARMLAND DATASET (%)

Dataset	Index	10	20	40	60	80
Farmland	OA	96.74	**97.13**	96.97	96.26	94.84
	Kappa	92.22	**93.05**	92.61	90.85	87.39
	Precision	92.03	94.81	**95.59**	94.80	92.11
	Recall	**97.19**	95.34	93.89	92.14	89.94
	F1-score	94.54	**95.07**	94.73	93.46	91.01

TABLE IV
RESULTS OF DIFFERENT SEGMENTATION SCALES ON THE SANTA BARBARA DATASET (%)

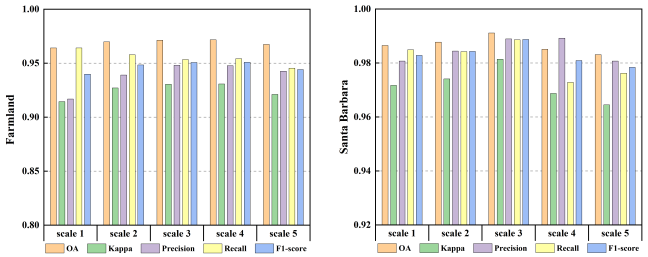
Dataset	Index	50	100	150	200	250
Santa Barbara	OA	98.88	98.32	**99.11**	98.69	98.65
	Kappa	97.66	96.50	**98.14**	97.26	97.17
	Precision	98.47	97.29	**98.89**	98.24	98.21
	Recall	98.69	98.48	**98.86**	98.44	98.36
	F1-score	98.58	97.88	**98.87**	98.34	98.28

The quantitative results of each algorithm on the Bay Area dataset are presented in Table I. The unsupervised algorithms obtain lower values than the semi-supervised algorithms for three composite metrics: OA, Kappa, and F1-score. MDGCN algorithm achieved the best prediction results among all the algorithms compared. Compared to MDGCN, our proposed algorithm shows improvements in five metrics by 0.57%, 1.13%, 0.85%, 0.21%, and 0.52%, respectively. From both qualitative and quantitative results, the ARCD method demonstrates significant superiority in the HSI CD task.

D. Ablation Study

In this section, we validate the effectiveness of the designed modules and discuss the effects of different superpixel segmentation scales, GCN scales, and training sample rates on the network. The Farmland and the Santa Barbara datasets are selected as distinct types of datasets for experiments.

1) Effectiveness of DAFF and PCAR Modules: To confirm the effectiveness of DAFF and PCAR, four ablation experiments are conducted, and the results are presented in Table II. The first experiment serves as the baseline for comparison. The integration of the DAFF module yields



(a) (b)

Fig. 12. Results of different DGCN scales on the Farmland and Santa Barbara datasets. (a) Results on the Farmland dataset. (b) Results on the Santa Barbara dataset.

significant improvements across all evaluation criteria on two datasets. This suggests that the DAFF module can effectively achieve the spectral–spatial augmentation information to supplement the difference information for improving the localization of the changed features. After replacing DIR with PCAR, the OA of the two datasets reached 96.75% and 98.84%, respectively. This indicates that the PCAR module can effectively reduce semantic gaps between different scale features of HSIs. Furthermore, the addition of both DAFF and PCAR modules resulted in the best experimental results on two datasets.

2) Discussion of Superpixel Segmentation Scales: Different superpixel segmentation scales have a significant impact on network performance. A smaller segmentation scale may result in inadequate segmentation of HSIs, while a larger segmentation scale may lead to excessive segmentation. Considering that EO-1 hyperion datasets contain concentrated changed regions, we set the experimental parameters for the superpixel segmentation scale to 10, 20, 40, 60, and 80. In contrast, the changed regions of the AVIRIS datasets are more fragmented, so we set the experimental parameters for the superpixel segmentation scale to be larger, specifically 50, 100, 150, 200, and 250. CD results of different superpixel segmentation scales on the Farmland dataset and the Santa Barbara dataset are presented in Tables III and IV, respectively. On two datasets, the network achieves optimal performance when the segmentation scales are set to 20 and 150, respectively. Therefore, for EO-1 hyperion datasets in this article, the superpixel segmentation scale is set to 20, while for AVIRIS datasets, the scale is set to 150.

3) Discussion of Different DGCN Scales: In our proposed method, different scales of DGCN can significantly affect the network performance. A small DGCN scale may prevent the network from fully capturing the multiscale features of HSIs, while a large scale may lead to over-smoothing. Therefore, we investigate five different DGCN scales and present the experimental results in Fig. 12. On the Farmland dataset, comparable performance is observed when the DGCN scale is set to 3 or 4. On the Santa Barbara dataset, all metrics reach their maximum values at a scale of 3. Based on the experimental results of two datasets and considering the complexity of the model, we have chosen a scale of 3 for DGCN.

4) Discussion of Training Sampling Rates: To further validate the robustness of ARCD, a series of experiments with varying training sample rates, specifically 0.2%, 0.5%, 1%,

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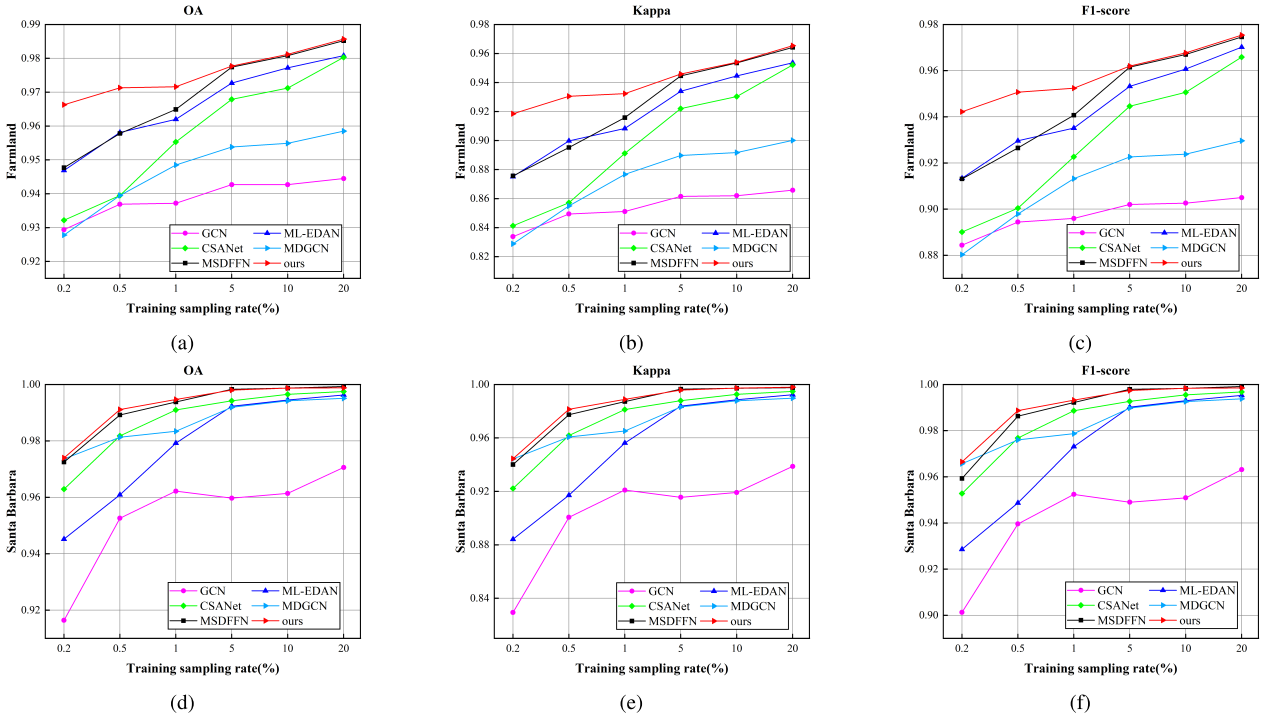


Fig. 13. OA, Kappa, and F1-score of different training sampling rates on the Farmland and Santa Barbara datasets. (a) OA on the Farmland dataset. (b) Kappa on the Farmland dataset. (c) F1-score on the Farmland dataset. (d) OA on the Santa Barbara dataset. (e) Kappa on the Santa Barbara dataset. (f) F1-score on the Santa Barbara dataset.

TABLE V
COMPUTATIONAL COST OF DIFFERENT METHODS ON THE SANTA BARBARA DATASET

Dataset	Index	CVA	SFA	RICD	GCN	ML_EDAN	CSANet	MDGCN	MSDFFN	ARCD
Santa Barbara	Params(K)	-	-	160312.1	19.9	112931.2	40459.9	33.7	26467.4	21.9
	Inference Times(s)	9.21	3.69	206.82	0.25	7.13	9.83	0.36	16.89	0.68

5%, 10%, and 20%, are conducted. OA, kappa, and F1-score metrics are selected for comparison. CD results of different training sample rates on the Farmland dataset and the Santa Barbara dataset are presented in Fig. 13. In both datasets, our proposed algorithm consistently achieves good results as the training sample rates increase. Notably, when the proportion of training sample rate is low, the superiority of our algorithm becomes even more evident, particularly on the Farmland dataset, where each metric value far outperforms those of other algorithms.

5) Discussion of Computational Cost: To investigate the computational cost of ARCD with other algorithms, the number of parameters (Params) and the inference times on the Santa Barbara dataset for different methods are measured with an NVIDIA GTX 3090Ti GPU, and the results are shown in Table V. It is obvious that graph-based methods (GCN, MDGCN, and ARCD) have fewer parameters and shorter inference times than other algorithms, which demonstrates the potential of GCN in terms of efficiency when dealing with hyperspectral data. Although ARCD did not achieve the least computational cost, such consumption is acceptable given that it has obtained the highest accuracy.

IV. CONCLUSION

In this article, a graph-based framework called ARCD, including the DMGCN subnetwork, DAFF module, and PCAR module is proposed for HSI CD. The DMGCN subnetwork focuses on the changed information of bi-temporal HSIs and learns multiscale information by capturing and integrating HSI features across multiple scales. The DAFF module fuses augmentation information and difference information for obtaining accurate difference features. Then, multilevel difference features are reconstructed by the PCAR module in a progressive manner. Our proposed ARCD not only improves the recognition capability of complex targets but also alleviates the demand for HSI labels. Quantitative and qualitative results on four HSI datasets show that ARCD outperforms several SOTA algorithms. For future work, we intend to develop an unsupervised graph-based DL approach to achieve performance comparable to semi-supervised methods, while eliminating the need for labels.

REFERENCES

[1] A. Singh, "Review article digital change detection techniques using remotely-sensed data," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 989–1003, Jun. 1989.

[2] S. Liu, M. Chi, Y. Zou, A. Samat, J. A. Benediktsson, and A. Plaza, "Oil spill detection via multitemporal optical remote sensing images: A change detection perspective," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 3, pp. 324–328, Mar. 2017.

[3] V. Amici, M. Marcantonio, N. La Porta, and D. Rocchini, "A multi-temporal approach in MaxEnt modelling: A new frontier for land use/land cover change detection," *Ecol. Informat.*, vol. 40, pp. 40–49, Jul. 2017.

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[4] X. Liu, H. Shen, Q. Yuan, X. Lu, and S. Li, "One-step high-quality NDVI time-series reconstruction by joint modeling of gradual vegetation change and negatively biased atmospheric contamination," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4407017.

[5] H. Gao, Z. Chen, and C. Li, "Shallow network based on depthwise overparameterized convolution for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[6] H. Fu et al., "Three-dimensional singular spectrum analysis for precise land cover classification from UAV-borne hyperspectral benchmark datasets," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 115–134, Sep. 2023.

[7] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[8] H. Fu, G. Sun, A. Zhang, B. Shao, J. Ren, and X. Jia, "Tensor singular spectral analysis for 3D feature extraction in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5403914.

[9] B. Yang, L. Qin, J. Liu, and X. Liu, "UTRNet: An unsupervised time-distance-guided convolutional recurrent network for change detection in irregularly collected images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410516.

[10] M. Hasanlou and S. T. Seydi, "Hyperspectral change detection: An experimental comparative study," *Int. J. Remote Sens.*, vol. 39, no. 20, pp. 7029–7083, Oct. 2018.

[11] A. Asokan and J. Anitha, "Change detection techniques for remote sensing applications: A survey," *Earth Sci. Informat.*, vol. 12, no. 2, pp. 143–160, Jun. 2019.

[12] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," in *Proc. LARS Symposia*, 1980, p. 385.

[13] F. Thonfeld, H. Feilhauer, M. Braun, and G. Menz, "Robust change vector analysis (RCVA) for multi-sensor very high resolution optical satellite data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 50, pp. 131–140, Aug. 2016.

[14] F. Bovolo, S. Marchesi, and L. Bruzzone, "A framework for automatic and unsupervised detection of multiple changes in multitemporal images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 6, pp. 2196–2212, Jun. 2012.

[15] Y. Afaq and A. Manocha, "Analysis on change detection techniques for remote sensing applications: A review," *Ecol. Informat.*, vol. 63, Jul. 2021, Art. no. 101310.

[16] A. P. Tewkesbury, A. J. Comber, N. J. Tate, A. Lamb, and P. F. Fisher, "A critical synthesis of remotely sensed optical image change detection techniques," *Remote Sens. Environ.*, vol. 160, pp. 1–14, Apr. 2015.

[17] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

[18] R. C. Gonzalez, *Digital Image Processing*. London, U.K.: Pearson, 2009.

[19] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2858–2874, May 2014.

[20] G. E. Hinton, S. Osindero, and Y.-W. Teh, "A fast learning algorithm for deep belief nets," *Neural Comput.*, vol. 18, no. 7, pp. 1527–1554, Jul. 2006.

[21] A. Aasbjerg Nielsen, "The regularized iteratively reweighted MAD method for change detection in Multi- and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.

[22] L. Wang, L. Wang, Q. Wang, and L. Bruzzone, "RSCNet: A residual self-calibrated network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5529917.

[23] M. Hearst, "Support vector machines," *IEEE Intell. Syst. Their Appl.*, vol. 13, no. 4, pp. 18–28, Jul./Aug. 1998.

[24] R. Mousazadeh, H. Ghaffarzadeh, J. Nouri, A. Gharagozlou, and M. Farahpour, "Land use change detection and impact assessment in anzali international coastal wetland using multi-temporal satellite images," *Environ. Monitor. Assessment*, vol. 187, no. 12, pp. 1–11, Dec. 2015.

[25] M. Volpi, D. Tuia, F. Bovolo, M. Kanevski, and L. Bruzzone, "Supervised change detection in VHR images using contextual information and support vector machines," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 20, pp. 77–85, Feb. 2013.

[26] C. Wu, B. Du, and L. Zhang, "A subspace-based change detection method for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 815–830, Apr. 2013.

[27] Y. Liu, J. Hu, X. Kang, J. Luo, and S. Fan, "Interactformer: Interactive transformer and CNN for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3183468.

[28] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.

[29] J. Li, J. Zhu, C. Li, X. Chen, and B. Yang, "CGTF: Convolution-guided transformer for infrared and visible image fusion," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–14, 2022.

[30] L. Jiao et al., "Graph representation learning meets computer vision: A survey," *IEEE Trans. Artif. Intell.*, vol. 4, no. 1, pp. 2–22, Feb. 2023.

[31] A. Voulodimos, N. Doulamis, A. Doulamis, and E. Protopapadakis, "Deep learning for computer vision: A brief review," *Comput. Intell. Neurosci.*, vol. 2018, pp. 1687–5265, Jan. 2018.

[32] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2019.

[33] J. Qu, S. Hou, W. Dong, Y. Li, and W. Xie, "A multilevel encoder–decoder attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518113.

[34] R. Song, W. Ni, W. Cheng, and X. Wang, "CSANet: Cross-temporal interaction symmetric attention network for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[35] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[36] B. Tu, X. Liao, Q. Li, Y. Peng, and A. Plaza, "Local semantic feature aggregation-based transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536115.

[37] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3095166.

[38] W. Dong, Y. Yang, J. Qu, S. Xiao, and Y. Li, "Local information-enhanced graph-transformer for hyperspectral image change detection with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509814.

[39] X. Wang, K. Zhao, X. Zhao, and S. Li, "TriTF: A triplet transformer framework based on parents and brother attention for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507213.

[40] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536814.

[41] X. Zhang, S. Tian, G. Wang, X. Tang, J. Feng, and L. Jiao, "CAST: A cascade spectral-aware transformer for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5520714.

[42] Z. Lv, H. Huang, W. Sun, M. Jia, J. A. Benediktsson, and F. Chen, "Iterative training sample augmentation for enhancing land cover change detection performance with deep learning neural net work," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, pp. 1–14, 2023, doi: 10.1109/TNNLS.2023.3282935.

[43] L. Zhang, M. Lan, J. Zhang, and D. Tao, "Stagewise unsupervised domain adaptation with adversarial self-training for road segmentation of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609413.

[44] L. Cai, Z. Wang, R. Kulathinal, S. Kumar, and S. Ji, "Deep low-shot learning for biological image classification and visualization from limited training samples," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 5, pp. 2528–2538, May 2023.

[45] G. Sun et al., "Large kernel spectral and spatial attention networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519915.

[46] J. Shi, Z. Zhang, T. Wu, X. Zhang, and Y. Lei, "Collaborative self-perception network architecture for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[47] J. Lei, M. Li, W. Xie, Y. Li, and X. Jia, "Spectral mapping with adversarial learning for unsupervised hyperspectral change detection," *Neurocomputing*, vol. 465, pp. 71–83, Nov. 2021.

[48] B. Yang, Y. Mao, L. Liu, X. Liu, Y. Ma, and J. Li, "From trained to untrained: A novel change detection framework using randomly initialized models with spatial–channel augmentation for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4402214.

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[49] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Toulon, France, Apr. 2017, pp. 1–14.
[50] M. Gong, H. Zhou, A. K. Qin, W. Liu, and Z. Zhao, "Self-paced co-training of graph neural networks for semi-supervised node classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9234–9247, Nov. 2023.
[51] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.
[52] J. Qu, Y. Xu, W. Dong, Y. Li, and Q. Du, "Dual-branch difference amplification graph convolutional network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519912.
[53] X. Wang, K. Zhao, X. Zhao, and S. Li, "CSDBF: Dual-branch framework based on temporal–spatial joint graph attention with complement strategy for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540118.
[54] T.-H. Lin and C.-H. Lin, "Hyperspectral change detection using semi-supervised graph neural network and convex deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515818.
[55] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.
[56] Y. Yang et al., "Semi-supervised multiscale dynamic graph convolution network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 5, pp. 6806–6820, May 2024.
[57] H. Xia, Y. Tian, L. Zhang, and S. Li, "A deep Siamese postclassification fusion network for semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622716.
[58] C. Zhang and B. Wang, "Progressive feature fusion framework based on graph convolutional network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3270–3284, 2024.
[59] H. Huang, Z. Chen, Y. Zou, M. Lu, and C. Chen, "Channel prior convolutional attention for medical image segmentation," 2023, *arXiv:2306.05196*.
[60] K. Xu, H. Huang, P. Deng, and Y. Li, "Deep feature aggregation framework driven by graph convolutional network for scene classification in remote sensing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5751–5765, Oct. 2022.
[61] Q. Guo, J. Zhang, C. Zhong, and Y. Zhang, "Change detection for hyperspectral images via convolutional sparse analysis and temporal spectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4417–4426, 2021.
[62] P. Henry, G. Chander, B. Fougnie, C. Thomas, and X. Xiong, "Assessment of spectral band impact on intercalibration over desert sites using simulation based on EO-1 hyperion data," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1297–1308, Mar. 2013.
[63] J. López-Fandiño, D. B. Heras, F. Argüello, and M. Dalla Mura, "GPU framework for change detection in multitemporal hyperspectral images," *Int. J. Parallel Program.*, vol. 47, no. 2, pp. 272–292, Apr. 2019.
[64] G. H. Rosenfield and K. Fitzpatrick-Lins, "A coefficient of agreement as a measure of thematic classification accuracy," *Photogramm. Eng. Remote Sens.*, vol. 52, no. 2, pp. 223–227, 1986.
[65] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 32, 2019, pp. 1–12.
[66] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, May 2015, pp. 1–6.



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